

## A BAYESIAN ESTIMATE FOR THE CONSTANTS IN RIDGE REGRESSION

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**Summary:** In the case of multicollinear data, ridge regression is considered as an alternative to least squares regression. Several authors have proposed estimates for the constants in generalized ridge regression. In this paper a Bayesian approach is adopted to estimate new constants. These constants are compared with those of other constants proposed in the literature. It is shown that the proposed estimates have advantages both with respect to bias and efficiency.

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### 1. INTRODUCTION

When multicollinearity is present amongst the independent or explanatory variables then least squares estimates may not be satisfactory because of high variability and its computational instability. Hoerle and Kennard (1970) introduced ridge regression in the statistical literature in order to obtain an estimator that dominates the ordinary least squares (ols) estimator in the presence of multicollinearity.

Consider the standard multiple regression model

$$(1) \quad Y = X\beta + e$$

where  $Y$  is an  $n$ -dimensional column vector of responses,  $X$  is a  $(n \times p)$  matrix of known constants and  $\beta$  is a  $p$ -dimensional column vector of regression coefficients and  $e$  follows a  $p$ -variate normal distribution with

mean vector 0 and variance covariance matrix  $\sigma^2 I$ , that is  $e \sim N(0, \sigma^2 I)$ .

It is common practice in ridge regression to standardize the  $X$  matrix so that  $X'X$  is in correlation form. We also standardize  $Y$  so that both  $X$  and  $Y$  are in the standardized form. This may have some scalar effect on  $\sigma^2$  but we simply assume that this scalar effect is incorporated in a new  $\sigma^2$ .

The ols estimate of  $\beta$  is given by

$$(1.2) \quad \hat{\beta} = (X'X)^{-1}X'Y$$

$$\text{and } \hat{\sigma}^2 = \frac{1}{n-p}(Y - X\hat{\beta})'(Y - X\hat{\beta})$$

When the matrix  $X$  is nearly multicollinear, the matrix  $X'X$  will be close to singular. The ridge estimator of  $\beta$  is defined by perturbing the matrix  $X'X$  away from singularity:

$$(1.3) \quad \hat{\beta}(k_i) = (X'X + K)^{-1}X'Y$$

where  $K = \text{diag}(k_1, \dots, k_p)$  and the  $k_i$  are called the ridge constants.

Let  $P$  be an orthogonal matrix such that  $P'X'XP = \Lambda = \text{diag}(\lambda_i)$ .

Then we rewrite model (1.1) as

$$(1.4) \quad Y = X\beta + e$$

$$= (XP)(P'\beta) + e$$

$$= Z\alpha + e$$

with

$$(1.5) \quad Z'Z = \Lambda$$

and

$$(1.6) \quad \hat{\alpha} = (Z'Z)^{-1}Z'Y$$

for the ols estimate of  $\alpha$  and

$$(1.7) \quad \hat{\alpha}(k_i) = (Z'Z + K)^{-1}Z'Y$$

for the ridge estimate.

Note that

$$(1.8) \quad \hat{\alpha}(k_i) = (Z'Z + K)^{-1}Z'Z\hat{\alpha}$$

$$= (\Lambda + K)^{-1}\Lambda\hat{\alpha}.$$

Thus

$$(1.9) \quad \hat{\alpha}_i(k_i) = (\lambda_i / (\lambda_i + k_i)) \hat{\alpha}_i$$

The mean square error of  $\hat{\alpha}(k_i)$  is given by

$$(1.10) \quad E(L_1^2(k_i)) = MSE(\hat{\alpha}(k_i)) = E(\hat{\alpha}(k_i) - \alpha)'(\hat{\alpha}(k_i) - \alpha) \\ = \sum_{i=1}^p \frac{\sigma^2 \lambda_i + \alpha_i^2 k_i^2}{(\lambda_i + k_i)^2}$$

Several authors minimize this function to obtain estimates for  $k_i$ . The Hoerl and Kennard estimates are

$$(1.11) \quad \hat{k}_i = \frac{\hat{\sigma}^2}{\hat{\alpha}_i^2}$$

while the Browne and Rock (1978) and Hemmerle and Brantle (1978) estimates are given by

$$(1.12) \quad \left. \begin{aligned} \hat{k}_i &= \frac{\lambda_i \hat{\sigma}^2}{\lambda_i \hat{\alpha}_i^2 - \hat{\sigma}^2} & \text{if } e_i < 1 \\ &= \infty & \text{if } e_i \geq 1 \end{aligned} \right\}$$

where

$$e_i = \frac{\hat{\sigma}^2}{\lambda_i \hat{\alpha}_i^2}$$

Note that if  $e \sim N(0, \sigma^2 I)$  then  $\hat{\alpha} \sim N(\alpha, \sigma^2 \lambda^{-1})$  and hence

$$(1.13) \quad \hat{\alpha} \sim N(\alpha_i, \sigma^2 / \lambda_i)$$

Thus the usual  $F$ -statistic to test the hypothesis  $H_0: \alpha_i = 0$  is

$$(1.14) \quad F_i = \frac{\lambda_i \hat{\alpha}_i^2}{\hat{\sigma}^2}$$

which is distributed as the  $F$ -statistic with 1 and  $n - p$  degrees of freedom.

Thus the estimate (1.12) reduces to

$$(1.15) \quad \hat{k}_i = \frac{\lambda_i}{F_i - 1} \quad \text{if } F_i > 1 \\ = \infty \quad \text{if } F_i \leq 1.$$

The estimate of  $k_i$  given by (1.15) is in a very convenient form. If  $k_i = k$  then the Lawless and Wang (1976) estimator for a single  $k$  is

$$(1.16) \quad \hat{k} = \frac{1}{F}$$

where  $F$  is the usual  $F$ -statistic to test  $H_0: \beta = 0$ .

The only drawback about estimator (1.15) is choosing  $\hat{k}_i = \infty$  if  $F_i \leq 1$ . In most cases, of course  $F_i$  will be larger than one, but cases have been reported by Browne and Rock (1978) where  $F_i \leq 1$ . This implies that  $\hat{\alpha}_i(\infty) = 0$ .

In the next section we derive an estimate for  $k_i$  that overcomes this problem.

## 2. A BAYESIAN ESTIMATE FOR THE RIDGE CONSTANTS

We assume the prior distribution of  $\alpha$  to be  $N(\alpha_0, \sigma^2 A^{-1})$  where  $A = \text{Diag}(\alpha_1, \dots, \alpha_p)$  while the prior distribution of  $\sigma$  is that of an inverted  $\chi^2$  distribution with  $\nu$  degrees of freedom, that is

$$(2.1) \quad p(\alpha, \sigma) = p(\alpha/\sigma)p(\sigma) \\ = \left[ \frac{1}{2\pi\sigma^2} \right]^{\frac{1}{2}p} |A|^{\frac{1}{2}} e^{-\frac{1}{2}(\alpha - \alpha_0)' A (\alpha - \alpha_0)/\sigma^2} \\ \cdot \left[ \frac{\nu}{2} s_0^2 \right]^{\frac{1}{2}\nu} e^{-\frac{\nu s_0^2}{2\sigma^2}} / \left( \frac{1}{2}\Gamma\left(\frac{1}{2}\nu\right) \right) \sigma^{-(\nu+1)}$$

This is the form of a natural conjugate prior distribution for  $(\alpha, \sigma)$ , where the conditional distribution of  $\alpha$  given  $\sigma$  is multivariate normal and the marginal distribution for  $\sigma$  is an inverted  $\chi^2_\nu$  distribution. The parameters  $\alpha_0$ ,  $A$ ,  $\nu$  and  $s_0^2$  depend on the prior information.

From Browne and Rock (1978) or Hemmerle and Brantle (1978),  $E(L_1^2(k))$  given by (1.10) can be written as

$$(2.2) \quad E(L_1^2(k_i)/\alpha, \sigma^2) = \sum_{i=1}^p \frac{k_i^2}{(\lambda_i + k_i)^2} (\sigma^2/\lambda_i + \alpha_i^2) + \sigma^2 \sum_{i=1}^p \frac{\lambda_i - k_i}{\lambda_i(\lambda_i + k_i)}$$

Minimization of (2.2) with respect to  $k_i$  would give solutions to the additive constants which involves population parameters. Before minimizing (2.2) we replace the population parameters by the known parameters of the natural conjugate prior distribution.

Now

$$(2.3) \quad E[(\sigma^2/\lambda_i + \alpha_i^2)/\sigma^2] = \sigma^2/\lambda_i + E(\alpha_i^2/\sigma^2)$$

but

$$E[(\alpha_i - \alpha_{i0})^2/\sigma^2] = \sigma^2/a_i \text{ since } E[(\alpha - \alpha_0)(\alpha - \alpha_0)'/\sigma^2] = \sigma^2 A^{-1}$$

so that

$$E(\alpha_i^2/\sigma^2) = \sigma^2/a_i + \alpha_{i0}^2.$$

Also

$$E(\sigma^2) = [\nu/(\nu - 2)]s_0^2 = s_0^{*2}.$$

Thus unconditionally

$$(2.4) \quad E(L_1^2(k_i)) = \sum_{i=1}^p \frac{k_i^2}{(\lambda_i + k_i)^2} (s_0^{*2}/\lambda_i + s_0^{*2}/a_i + \alpha_{i0}^2) + s_0^{*2} \sum_{i=1}^p \frac{\lambda_i - k_i}{\lambda_i(\lambda_i + k_i)}.$$

We will now minimize this function with respect to  $k_i$ .

The  $i$ -th element of (2.4) is

$$(2.5) \quad M_i = \frac{k_i^2}{(\lambda_i + k_i)^2} B + \frac{s_0^{*2}}{\lambda_i} \cdot \frac{\lambda_i - k_i}{\lambda_i + k_i}$$

where

$$(2.6) \quad B = (s_0^{*2}/\lambda_i + s_0^{*2}/a_i + \alpha_{i0}^2)$$

Let

$$(2.7) \quad \nu_i = \frac{\lambda_i}{\lambda_i + k_i}$$

then

$$\begin{aligned} (\nu_i - 1)^2 &= k_i^2/(\lambda_i + k_i)^2 \text{ and} \\ \nu_i^2 - (\nu_i - 1)^2 &= (\lambda_i - k_i)/(\lambda_i + k_i). \end{aligned}$$

Thus substituting in (2.5) we have

$$\begin{aligned} M_i &= (\nu_i - 1)^2 B + \frac{s_0^{*2}}{\lambda_i} (2\nu_i - 1) \\ &= B[(\nu_i - 1)^2 + 2(\nu_i - 1)s_0^{*2}/\lambda_i B] + s_0^{*2}/\lambda_i B \\ &= B[(\nu_i - 1 + s_0^{*2}/\lambda_i B)^2 - (s_0^{*2}/\lambda_i B)^2] + s_0^{*2}/\lambda_i B \end{aligned}$$

so that

$$\frac{\partial M_i}{\partial \nu_i} = 2B[(\nu_i - 1 + s_0^{*2}/\lambda_i B)] = 0$$

yielding

$$\begin{aligned} \nu_i &= 1 - s_0^{*2}/\lambda_i B & \text{if } s_0^{*2}/\lambda_i B < 1 \\ &= 0 & \text{if } s_0^{*2}/\lambda_i B \geq 1. \end{aligned}$$

For  $k_i$  we get

$$(2.8) \quad k_i = \frac{a_i}{\frac{a_i \alpha_{i0}^2}{s_0^{*2}} + 1}$$

if  $s_0^{*2}/\lambda_i B < 1$ . Thus of crucial importance is knowledge of the prior information since the parameters  $\alpha$ ,  $A$ ,  $\nu$  and  $s_0^2$  are given by the conjugate

prior distribution.

Since  $\alpha = P'\beta$  with  $P$  known, we need only be concerned with the natural conjugate prior of  $(\beta, \sigma)$ . Assuming that the conjugate prior of  $p(\beta/\sigma)$  is multivariate normal  $N(\beta_0, \sigma^2 \bar{A}^{-1})$  and that  $p(\sigma)$  has an inverted  $\chi_\nu^2$  distribution (see 2.1), then (Judge et al. (1985), p. 107)

$$E(\beta/\sigma) = E(\beta) = \beta_0, \text{cov}(\beta/\sigma) = \sigma^2 \bar{A}^{-1}$$

and

$$E(\sigma^2) = [\nu/(\nu - 2)]s_0^2 = s_0^{*2}$$

Using Bayes' theorem the joint posterior density of  $(\beta, \sigma)$  given the data  $Y$  is

$$g(\beta, \sigma/Y) \propto l(\beta, \sigma/Y)g(\beta, \sigma)$$

where  $l(\beta, \sigma/Y)$  is the likelihood and  $g(\beta, \sigma) = p(\beta/\sigma)p(\sigma)$  given above. The joint posterior density is given in Judge et al. (1985), p. 108, from which it follows that

$$E(\beta/Y) = \tilde{\beta} = (\bar{A} + X'X)^{-1}(\bar{A}\beta_0 + X'X\hat{\beta}),$$

$$\text{cov}(\beta/Y) = \bar{s}^2(\bar{A} + X'X)^{-1}$$

where

$$\bar{\nu}\bar{s}^2 = \nu s_0^2 + Y'Y + \beta_0'\bar{A}\beta_0 - \tilde{\beta}'(\bar{A} + X'X)\tilde{\beta},$$

and

$$E(\sigma^2) = [\bar{\nu}/(\bar{\nu} - 2)]\bar{s}^2, \quad \bar{\nu} = n + \nu.$$

If  $L(\theta, \hat{\theta})$  is the loss incurred, then if a decision is based on the point  $\hat{\theta}$ , when  $\theta$  is the true parameter then the Bayes estimate for  $\theta$  is that value  $\hat{\theta}$  that minimizes

$$E_{\theta/Y}(L(\theta, \hat{\theta})) = \int L(\theta, \hat{\theta})g(\theta/Y)d\theta$$

where  $g(\theta/Y)$  is the posterior of  $\theta$  given the data  $Y$ . For the general quadratic loss function

$$L(\theta, \hat{\theta}) = (\theta - \hat{\theta})'Q(\theta - \hat{\theta})$$

where  $Q$  is a positive definite matrix, the Bayes estimate is the mean of the posterior distribution. Thus the Bayes estimators under quadratic loss are

$$\hat{\beta}_{\text{Bayes}} = (\bar{A} + X'X)^{-1}(\bar{A}\beta_0 + X'X\hat{\beta})$$

and

$$\hat{\sigma}_{Bayes}^2 = [\bar{\nu}/(\nu - 2)]\bar{s}^2$$

Thus

$$\hat{\alpha}_{Bayes} = P' \hat{\beta}_{Bayes}$$

In terms of  $k_i$  we have

$$(2.9) \quad \hat{k}_i = \frac{\bar{a}_i}{\frac{\bar{a}_i \hat{\alpha}_i^2 (Bayes)}{\hat{\sigma}_{Bayes}^2} + 1}$$

where  $\bar{a}_i$  is the  $i$ -th diagonal element of  $(\bar{A} + X'X)^{-1}$ .

It is worth noting that using a natural conjugate prior can be considered equivalent to having actual information from a previous sample, say  $(Y_1, X_1)$  and the present sample is  $(Y, X)$ . The analysis follows in exactly the same way as that of a natural conjugate prior (see Zellner (1971), pp. 70-72). In this case  $n_1$  is the sample size and  $s_1^2$  the estimate for  $\sigma^2$ ,

$$\bar{A} = X_1' X_1, X_1' Y_1 = \bar{A} \beta_0, s_1^2 = s_0^2 \text{ and } \nu = n_1 - p$$

For comparison with the Browne and Rock (1978) estimates for  $k_i$  we have decided to use the same examples, namely that of Gorman and Toman (1966) and Jeffers (1967).

Since a natural conjugate prior nor a previous sample is available, we decided to use the estimates given by the present sample, that is  $s^2$  for  $\hat{\sigma}_{Bayes}^2$ ,  $\hat{\alpha} = P' \hat{\beta}$  for  $\hat{\alpha}_{Bayes}$  and the constant  $\bar{a}_i$  is replaced by  $\lambda_i$ . Thus

$$(2.10) \quad \hat{k}_i = \frac{\lambda_i}{\frac{\lambda_i \hat{\alpha}_i^2}{s^2} + 1} = \frac{\lambda_i}{F_i + 1}$$

where  $F_i = \frac{\lambda_i \hat{\alpha}_i^2}{s^2}$  and is the  $F$ -statistic to test  $H_0: \alpha_i = 0$ .

Now (2.10) holds for (see (2.8))

$$s_0^2 / \lambda_i B < 1$$

or

$$s_0^2/\lambda_i \left[ \frac{s_0^2}{\lambda_i} + \frac{s_0^2}{a_i} + \alpha_{i0}^2 \right] < 1.$$

If we replace  $s_0^2$  by  $s^2$ ,  $a_i$  by  $\lambda_i$  and  $\alpha_{i0}$  by  $\hat{\alpha}_i$ , we have

$$\begin{aligned} s^2/\lambda_i \left[ s^2 \left[ \frac{1}{\lambda_i} + \frac{1}{\lambda_i} \right] + \hat{\alpha}_i^2 \right] &< 1 \\ &= s^2/(2s^2 + \lambda_i \hat{\alpha}_i^2) < 1 \\ &= 1/\left[ 2 + \frac{\lambda_i \hat{\alpha}_i^2}{s^2} \right] < 1 \\ &= 1/(2 + F_i) < 1. \end{aligned}$$

or

$$F_i > -1.$$

Since  $F_i \geq 0$  above will always hold and we have

$$(2.11) \quad \hat{k}_i = \lambda_i/(F_i + 1).$$

Note that the Browne and Rock estimate is

$$(2.12) \quad \begin{aligned} \hat{k}_i &= \lambda_i/(F_i - 1) && \text{for } F_i > 1 \\ &= \infty && \text{for } F_i \leq 1 \end{aligned}$$

For  $F_i$  large both (2.11) and (2.12) will be close. For small  $F_i$  (2.11) will be close to  $\lambda_i$ . Of course  $\lambda_i$  may also be large but can never be larger than 1, and at least some exact value is available. Since  $\sum \lambda_i = p$  an average value for  $\lambda_i$  will be  $1/p$ . We therefore propose that in all cases, where prior information is not available, that (2.11) be used as an estimate for  $k_i$ .

### 3. EXAMPLES

We present the two examples that are given in Browne and Rock (1978), namely the ten factor regression model of Gorman and Toman (1966) and the thirteen factor problem of Jeffers (1967). The correlation matrices given by Gorman and Toman and Jeffers were employed. All the coefficients were recomputed and all estimates agree with those given by Browne and Rock (1978). The additive constants and regression weights are given in tables 1 and 2.

TABLE 1. Regression weights and additive constants for Gorman and Toman's example  $n = 36$ . Condition number equals 54.3

| Browne and Rock |          |                 | Troskie and Chalton |                 |        |
|-----------------|----------|-----------------|---------------------|-----------------|--------|
| $i$             | $k_i$    | $\hat{\beta}_i$ | $k_i$               | $\hat{\beta}_i$ | $F_i$  |
| 1               | 0.0297   | -0.148          | 0.0292              | -0.161          | 125.41 |
| 2               | 0.0404   | -0.162          | 0.0384              | -0.191          | 39.16  |
| 3               | $\infty$ | -0.312          | 0.9321              | -0.327          | 0.3869 |
| 4               | 0.0894   | -0.061          | 0.0764              | -0.088          | 12.68  |
| 5               | 12.7230  | -0.348          | 0.4682              | -0.391          | 1.076  |
| 6               | 0.0623   | 0.785           | 0.0524              | 0.781           | 11.56  |
| 7               | 0.0643   | 0.255           | 0.0473              | 0.253           | 6.56   |
| 8               | 0.0497   | 0.369           | 0.0342              | 0.361           | 5.42   |
| 9               | 0.0851   | 0.091           | 0.0401              | 0.089           | 2.78   |
| 10              | 0.0062   | 0.070           | 0.0053              | 0.085           | 11.91  |

TABLE 2. Jeffers' example  $n = 180$ . Condition number equals 108.2

| Browne and Rock |          |                 | Troskie and Chalton |                 |         |
|-----------------|----------|-----------------|---------------------|-----------------|---------|
| $i$             | $k_i$    | $\hat{\beta}_i$ | $k_i$               | $\hat{\beta}_i$ | $F_i$   |
| 1               | 0.0925   | -0.483          | 0.0886              | -0.477          | 46.61   |
| 2               | 0.0116   | 0.366           | 0.0115              | 0.378           | 205.21  |
| 3               | 0.1053   | -0.934          | 0.0947              | -0.946          | 18.84   |
| 4               | 1.9336   | 0.272           | 0.4310              | 0.274           | 1.57    |
| 5               | 0.0109   | -0.084          | 0.0106              | -0.081          | 84.48   |
| 6               | 0.0221   | 0.126           | 0.0210              | 0.156           | 37.84   |
| 7               | 0.0304   | 0.220           | 0.0275              | 0.167           | 19.97   |
| 8               | 0.0639   | -0.221          | 0.0495              | -0.237          | 7.88    |
| 9               | $\infty$ | -0.070          | 0.2906              | -0.073          | 0.2136  |
| 10              | $\infty$ | -0.132          | 0.1908              | -0.102          | 0.00040 |
| 11              | $\infty$ | 0.107           | 0.0357              | 0.099           | 0.4183  |
| 12              | 0.0015   | -0.066          | 0.0014              | -0.064          | 27.80   |
| 13              | $\infty$ | 0.008           | 0.0387              | 0.003           | 0.00008 |

Although the two estimates appear to be not that different the instability of the Browne and Rock estimates for  $F_i$  values close to or less than 1 are clearly demonstrated.

## 4. MOMENTS OF THE ESTIMATOR

The moments are obtained following the approach of Dwivedi et al. (1980).

From (2.11) we have

$$(4.1) \quad \hat{k}_i = \lambda_i / (F_i + 1) = \frac{s^2 \lambda_i}{\lambda_i \hat{\alpha}_i^2 + s^2}$$

Thus

$$(4.2) \quad \hat{\alpha}_i(\hat{k}_i) = \left[ \frac{1}{\lambda_i + \frac{s^2 \lambda_i}{\lambda_i \hat{\alpha}_i^2 + s^2}} \right] z_i' y$$

$$= \left[ \frac{\lambda_i \hat{\alpha}_i^2 + s^2}{\lambda_i^2 \hat{\alpha}_i^2 + 2s^2 \lambda_i} \right] z_i' y$$

where  $z_i$  is the  $i$ -th column of  $Z$ .

Following some straightforward calculations and using the results on expectations given by Dwivedi et al. (1980, p. 210), we obtain the following theorem:

**THEOREM.** The first and second moments of the estimator in (4.2) are given by

$$(4.3) \quad E(\hat{\alpha}_i(\hat{k}_i)) = \alpha_i \frac{\exp(-\tau_i^2/2)}{\Gamma(f/2)} \sum_{k=0}^{\infty} \left[ \frac{f/2 - 1}{f/2} \right]^k \sum_{j=0}^{\infty} \frac{(\tau_i^2/2)^j}{j!} \Gamma(j + (f+1)/2 + 1)$$

$$\times \left[ \frac{\Gamma(k + f/2)}{\Gamma(j + k + (f+1)/2 + 1)} - (1 - 1/f) \frac{\Gamma(k + f/2 + 1)}{\Gamma(j + k + (f+3)/2 + 1)} \right]$$

and

$$\begin{aligned}
 (4.4) \quad E((\hat{\alpha}_i(\hat{k}_i))^2) &= \alpha_i^2 \frac{\exp(-\tau_i^2/2)}{\Gamma(f/2)} \sum_{k=0}^{\infty} (k+1) \left[ \frac{f/2-1}{f/2} \right]^k \\
 &\times \sum_{j=0}^{\infty} \frac{(\tau_i^2/2)^{j-1}}{j!} \frac{\Gamma(j+(f+3)/2)\Gamma(j+3/2)}{\Gamma(j+1/2)} \\
 &\times \left[ \frac{\Gamma(k+f/2)}{\Gamma(j+k+(f+3)/2)} - 2(1-1/f) \frac{\Gamma(k+f/2+1)}{\Gamma(j+k+(f+3)/2+1)} \right. \\
 &\left. + (1-1/f)^2 \frac{\Gamma(k+f/2+2)}{\Gamma(j+k+(f+3)/2+2)} \right]
 \end{aligned}$$

where  $\tau_i^2$  is the noncentrality parameter given by

$$\tau_i^2 = \lambda_i \alpha_i^2 / \sigma^2$$

## 5. A COMPARISON OF GENERALIZED RIDGE ESTIMATORS

A number of generalized ridge estimators have been compared by Hemmerle and Carey (1983) and Noor and Mehta (1989). In addition to the estimators using the estimates of  $k_i$  given by (1.11) and (1.12), labelled E0 and E3, respectively, by Hemmerle and Carey (1983), these authors also considered the estimator obtained by Hemmerle (1975) as the closed form solution to the iterative estimation procedure proposed by Hoerl and Kennard (1970). This estimator, which they denoted by E1, is given by

$$\begin{aligned}
 (5.1) \quad \hat{\alpha}_i^* &= 0 & \hat{\tau}_i^2 &\leq 4 \\
 &= \left( 1/2 + \sqrt{(1/4 - 1/\hat{\tau}_i^2)} \right) \hat{\alpha}_i & \hat{\tau}_i^2 &\geq 4
 \end{aligned}$$

where  $\hat{\tau}_i^2$  is the  $F$  statistic given in (1.14).

The proposed estimator, using (2.11) for the estimator of  $k_i$ , will be compared with the above three estimators, and, using the notation of Hemmerle and Carey (1983), it is convenient to denote the proposed estimator by E2.

It is noted that Noor and Mehta (1989) have introduced a further ridge estimator given by

$$(1 - a \exp(-b\hat{\tau}_i^2))\hat{\alpha}_i$$

which incorporates two arbitrary constants  $a$  and  $b$ . In practice this estimator has limitations as the optimal values of  $a$  and  $b$  have to be found by trial and error. In view of such limitation the estimator has not been included in the comparison of estimators, though it is acknowledged that using optimal values of  $a$  and  $b$  the estimator is superior to  $E0$ ,  $E1$  and  $E3$  in certain regions of the parameter space, as shown by Noor and Mehta (1989).

For the comparison of generalized ridge estimators the criteria to be used are the Relative Bias, given by

$$RB(\hat{\alpha}_i(\hat{k}_i)) = E\left[\frac{\hat{\alpha}_i(\hat{k}_i) - \alpha_i}{\alpha_i}\right]$$

and the Relative Efficiency (by comparison with the least squares estimator),

$$RE(\hat{\alpha}_i(\hat{k}_i)) = \left[ \frac{E(\hat{\alpha}_i(\hat{k}_i) - \alpha_i)^2}{E(\hat{\alpha}_i - \alpha_i)^2} \right] \times 100.$$

The estimators are

$$(5.2) \quad E0 = \left[ \frac{\hat{\tau}_i^2}{\hat{\tau}_i^2 + 1} \right] \hat{\alpha}_i$$

$$E1 = 0 \quad \hat{\tau}_i^2 \leq 4$$

$$= (1/2 + \sqrt{1/4 - 1/\hat{\tau}_i^2}) \hat{\alpha}_i \quad \hat{\tau}_i^2 \geq 4$$

$$E2 = \left[ \frac{\hat{\tau}_i^2 + 1}{\hat{\tau}_i^2 + 2} \right] \hat{\alpha}_i$$

$$E3 = 0 \quad \hat{\tau}_i^2 \leq 1$$

$$= (1 - 1/\hat{\tau}_i^2) \hat{\alpha}_i \quad \hat{\tau}_i^2 \geq 1$$

Results for estimators  $E0$ ,  $E1$  and  $E3$  have been extracted from Hemmerle and Carey (1983) and these together with the results for  $E2$  are shown in tables 3 and 4. Figures I and II illustrate the behaviour of the relative bias and relative efficiency for the four estimators when the number of degrees of freedom is 10.

TABLE 3. Relative efficiencies with respect to the least squares estimator.

|       |    | V      |        |        |        |        |
|-------|----|--------|--------|--------|--------|--------|
|       |    | 5      | 10     | 20     | 40     | 100    |
| 0.01  | E2 | 62.07  | 60.51  | 59.70  | 59.19  | 54.68  |
|       | E0 | 51.70  | 49.67  | 48.53  | 46.80  | 37.10  |
|       | E3 | 40.57  | 37.76  | 36.25  | 35.46  | 34.98  |
|       | E1 | 26.46  | 22.25  | 19.90  | 18.66  | 17.90  |
| 0.10  | E2 | 65.75  | 64.34  | 63.60  | 63.13  | 59.02  |
|       | E0 | 57.96  | 56.17  | 55.17  | 53.66  | 44.89  |
|       | E3 | 50.96  | 48.63  | 47.37  | 46.72  | 46.32  |
|       | E1 | 42.47  | 39.28  | 37.51  | 36.58  | 36.00  |
| 0.50  | E2 | 79.42  | 78.55  | 78.08  | 77.78  | 75.11  |
|       | E0 | 80.39  | 79.46  | 78.94  | 78.22  | 72.82  |
|       | E3 | 87.97  | 87.18  | 86.74  | 86.51  | 86.37  |
|       | E1 | 103.74 | 104.48 | 104.97 | 105.25 | 105.43 |
| 0.70  | E2 | 84.87  | 84.21  | 83.85  | 83.62  | 81.49  |
|       | E0 | 88.91  | 88.28  | 87.94  | 87.50  | 83.38  |
|       | E3 | 101.80 | 101.47 | 101.28 | 101.18 | 101.12 |
|       | E1 | 128.83 | 131.08 | 132.44 | 133.19 | 133.67 |
| 1.00  | E2 | 91.63  | 91.25  | 91.04  | 90.90  | 89.38  |
|       | E0 | 99.06  | 98.78  | 98.64  | 98.51  | 95.90  |
|       | E3 | 117.93 | 118.04 | 118.08 | 118.10 | 118.10 |
|       | E1 | 160.36 | 164.26 | 166.55 | 167.80 | 168.60 |
| 2.00  | E2 | 105.65 | 105.83 | 105.91 | 105.94 | 105.51 |
|       | E0 | 117.68 | 117.94 | 118.10 | 118.40 | 118.33 |
|       | E3 | 144.83 | 145.04 | 145.11 | 145.14 | 145.15 |
|       | E1 | 222.99 | 227.72 | 230.26 | 231.56 | 232.36 |
| 5.00  | E2 | 115.17 | 115.51 | 115.69 | 115.79 | 115.78 |
|       | E0 | 123.60 | 123.64 | 123.63 | 123.78 | 123.72 |
|       | E3 | 140.99 | 139.70 | 139.00 | 138.64 | 138.40 |
|       | E1 | 220.47 | 212.92 | 208.11 | 205.40 | 203.68 |
| 20.00 | E2 | 107.38 | 107.17 | 107.06 | 107.00 | 106.97 |
|       | E0 | 108.21 | 107.88 | 107.69 | 107.64 | 107.38 |
|       | E3 | 109.23 | 108.68 | 108.40 | 108.25 | 108.13 |
|       | E1 | 110.96 | 109.84 | 109.36 | 109.14 | 109.01 |

TABLE 4. Relative bias of the estimators

| $\tau^2/2$ |    | V      |        |        |        |        |
|------------|----|--------|--------|--------|--------|--------|
|            |    | 5      | 10     | 20     | 40     | 100    |
| 0.01       | E2 | -0.223 | -0.231 | -0.237 | -0.239 | -0.256 |
|            | E0 | -0.318 | -0.330 | -0.337 | -0.346 | -0.406 |
|            | E3 | -0.471 | -0.491 | -0.502 | -0.508 | -0.512 |
|            | E1 | -0.702 | -0.746 | -0.771 | -0.785 | -0.793 |
| 0.10       | E2 | -0.217 | -0.226 | -0.230 | -0.233 | -0.249 |
|            | E0 | -0.307 | -0.319 | -0.326 | -0.335 | -0.392 |
|            | E3 | -0.455 | -0.474 | -0.484 | -0.490 | -0.493 |
|            | E1 | -0.684 | -0.727 | -0.751 | -0.764 | -0.773 |
| 0.50       | E2 | -0.194 | -0.202 | -0.206 | -0.208 | -0.221 |
|            | E0 | -0.268 | -0.278 | -0.283 | -0.290 | -0.339 |
|            | E3 | -0.391 | -0.406 | -0.414 | -0.418 | -0.421 |
|            | E1 | -0.609 | -0.645 | -0.666 | -0.677 | -0.684 |
| 0.70       | E2 | -0.184 | -0.191 | -0.195 | -0.198 | -0.209 |
|            | E0 | -0.251 | -0.260 | -0.265 | -0.271 | -0.315 |
|            | E3 | -0.364 | -0.377 | -0.384 | -0.387 | -0.389 |
|            | E1 | -0.575 | -0.607 | -0.626 | -0.636 | -0.642 |
| 1.00       | E2 | -0.170 | -0.177 | -0.181 | -0.183 | -0.192 |
|            | E0 | -0.228 | -0.236 | -0.241 | -0.246 | -0.284 |
|            | E3 | -0.327 | -0.337 | -0.343 | -0.346 | -0.348 |
|            | E1 | -0.527 | -0.554 | -0.570 | -0.578 | -0.583 |
| 2.00       | E2 | -0.135 | -0.140 | -0.143 | -0.144 | -0.149 |
|            | E0 | -0.172 | -0.177 | -0.180 | -0.183 | -0.207 |
|            | E3 | -0.235 | -0.240 | -0.243 | -0.245 | -0.245 |
|            | E1 | -0.393 | -0.406 | -0.413 | -0.416 | -0.418 |
| 5.00       | E2 | -0.079 | -0.081 | -0.083 | -0.083 | -0.084 |
|            | E0 | -0.091 | -0.093 | -0.094 | -0.095 | -0.100 |
|            | E3 | -0.110 | -0.111 | -0.111 | -0.111 | -0.111 |
|            | E1 | -0.172 | -0.167 | -0.164 | -0.163 | -0.162 |
| 20.00      | E2 | -0.024 | -0.024 | -0.024 | -0.024 | -0.024 |
|            | E0 | -0.025 | -0.025 | -0.025 | -0.025 | -0.025 |
|            | E3 | -0.026 | -0.026 | -0.026 | -0.026 | -0.026 |
|            | E1 | -0.027 | -0.027 | -0.027 | -0.027 | -0.027 |

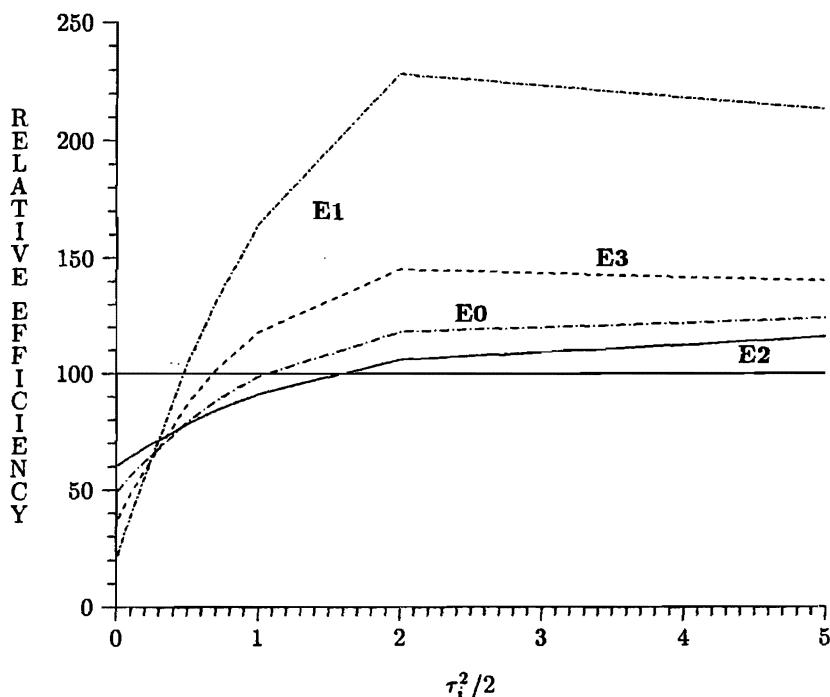


FIGURE I. Relative efficiency of the estimators with respect to least squares

In their conclusion of these estimators, Hemmerle and Carey (1983) came to the conclusion that E0 showed overall superiority compared with E1 and E3. The evidence from tables 3 and 4 and figures I and II is that the proposed estimator E2 shows similar superiority over E0, E1 and E3, and the same conclusions can be drawn viz.:

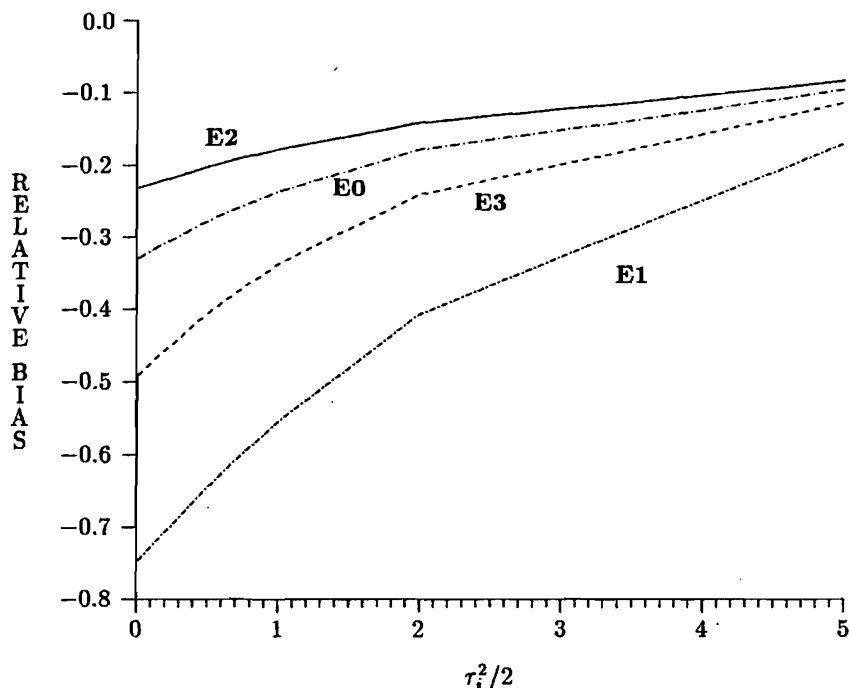


FIGURE II. Relative bias of the estimators with respect to  $\tau_i^2/2$

- (a) E2 has uniformly less bias than E0, E1 and E3:
- (b) E2 has uniformly a smaller maximum relative efficiency than E0, E1 and E3 with respect to least squares:
- (c) E2 is more efficient than least squares for larger values of  $\tau_i^2$  than is E0, E1 or E3:
- (d) The potential gain in efficiency for small  $\tau_i^2$  values, using E0, E1 or E3 as opposed to E2, is relatively small.

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